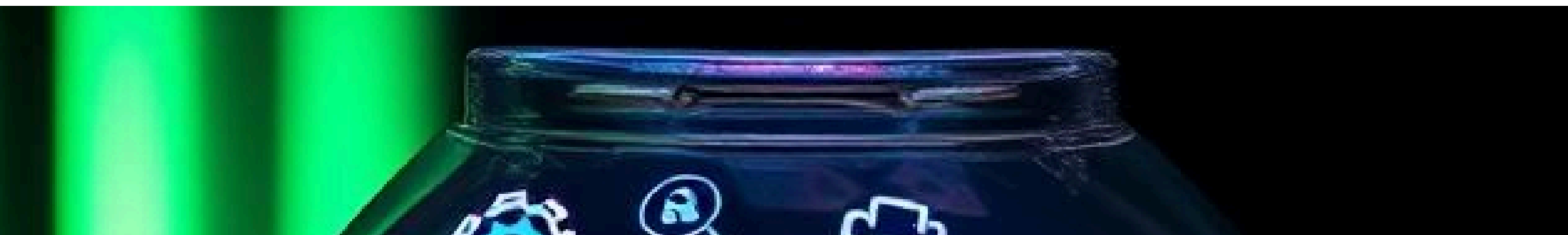




sos report

Ever troubleshooted a Linux system and wished for
a quick way to gather all the diagnostic info?
Meet the sos command!

Prepared By: Hosni Zaaraoui

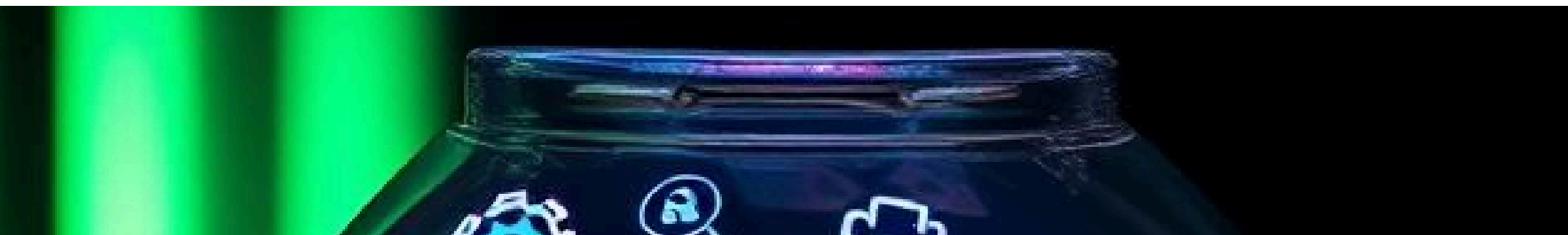


What is sos?

The `sos` command is a powerful diagnostic tool in Linux designed to collect system configuration, logs, and performance data into a single compressed archive.

Widely used by system administrators and support teams, it streamlines troubleshooting for distributions like Red Hat, Ubuntu, and CentOS. Its primary output, the `sosreport`, is essential for diagnosing issues efficiently.





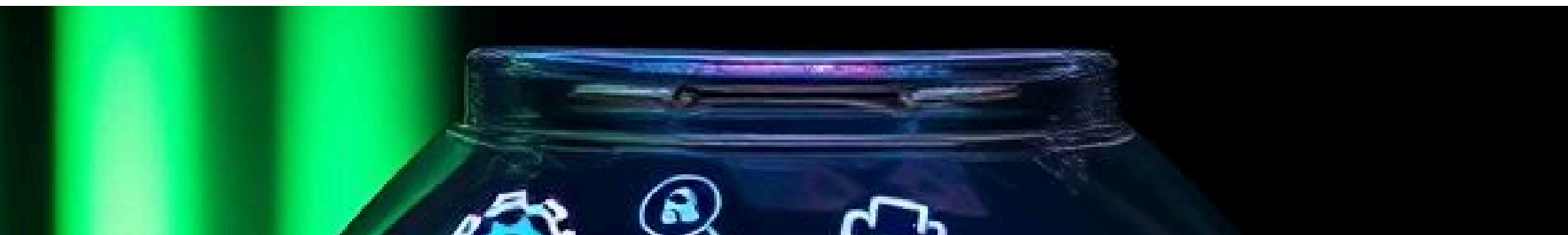
Key Features

Collects system logs (/var/log), hardware details, and network configurations.

Generates a .tar.xz file for easy sharing with support teams.

Supports modular plugins for customized data collection





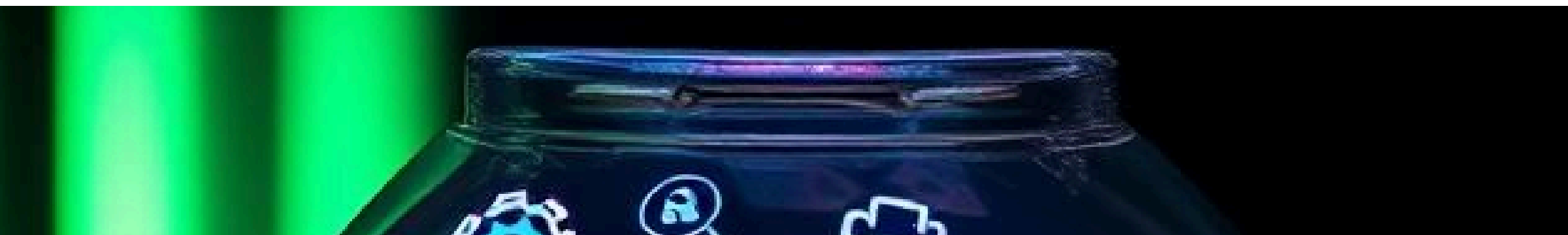
From sosreport to sos report

As of the sos-4.8 release, the legacy sosreport command has been deprecated in favor of sos report.

This change aligns with the tool's modular redesign, with sosreport set to be removed in the upcoming sos-4.9 release.

Users running `sudo sosreport` will see a warning and automatic redirection to `sos report`.

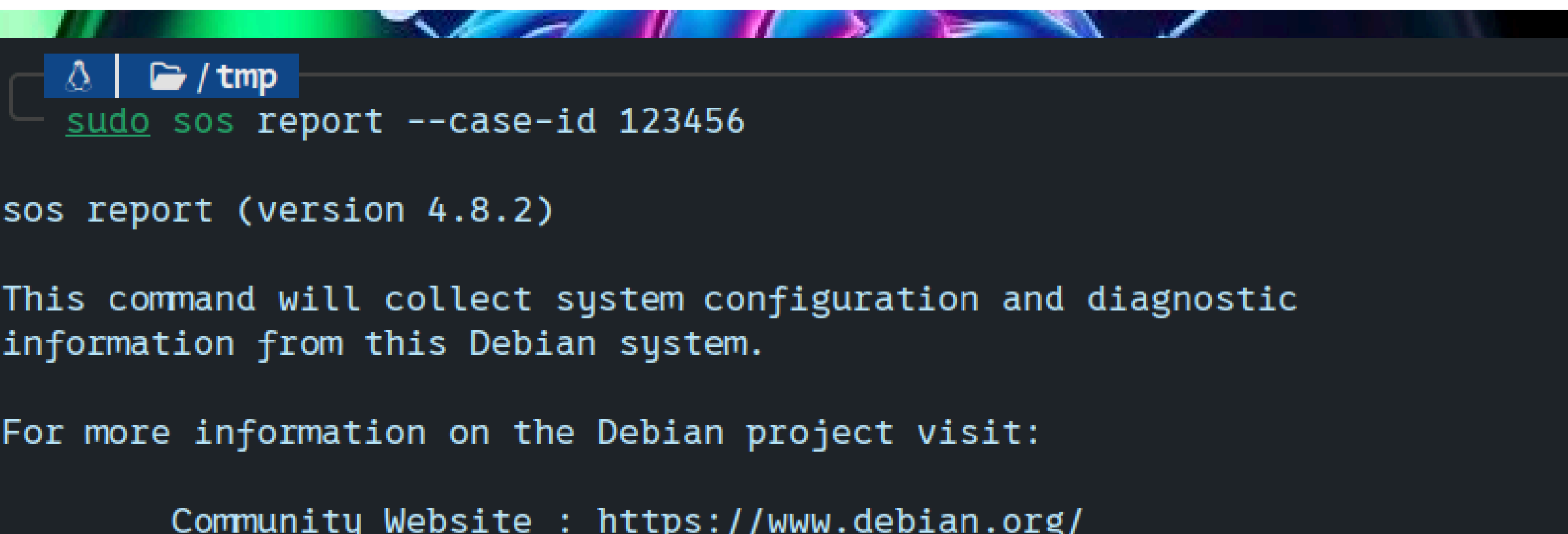




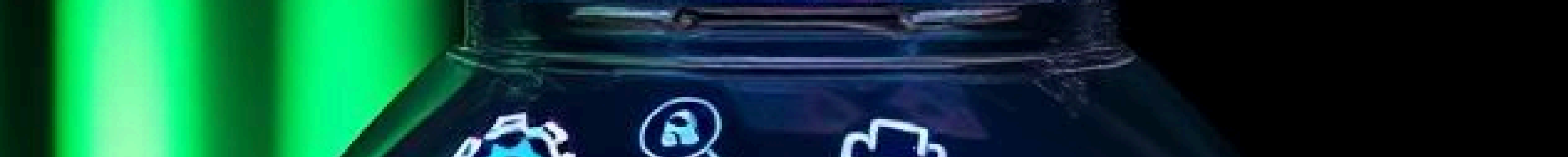
Running sos report

The sos report command is executed with superuser privileges to gather comprehensive system data. Here's how to use it:

1. Run **sudo sos report** in the terminal.
2. Respond to prompts (e.g., support case number, optional plugins).
3. The tool collects data and saves it as a .tar.xz file in /tmp (e.g., sosreport-hostname-2025.tar.xz).



```
🔍 | 📁 /tmp  
sudo sos report --case-id 123456  
  
sos report (version 4.8.2)  
  
This command will collect system configuration and diagnostic  
information from this Debian system.  
  
For more information on the Debian project visit:  
  
Community Website : https://www.debian.org/
```



Advanced Features and Customization

The sos tool is highly customizable through its plugin-based architecture.

Key options include:

Plugin Selection: Enable specific plugins with `-enable-plugins=plugin1,plugin2` or disable with `-skip-plugins=plugin1`.

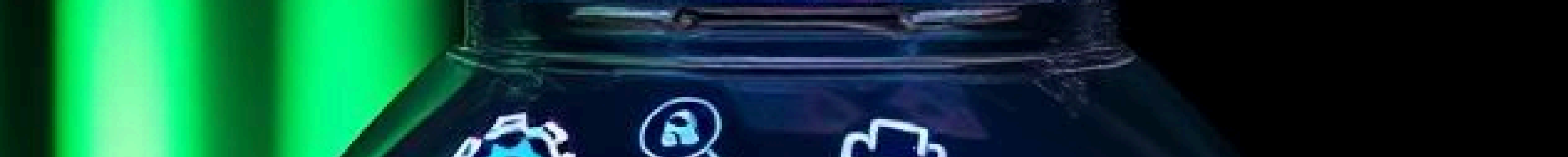
Compression Control: Use `-compression-type` to select `gzip`, `xz`, or `bzip2`.

Batch Mode: Run `sos report -batch` for non-interactive execution, ideal for automation



```
Running plugins. Please wait ...
```

```
Starting 2/74  anacron      [Running: anacron]
Starting 1/74  acpid       [Running: anacron acpid]
Starting 4/74  apt         [Running: anacron apt]
Starting 5/74  ata         [Running: anacron apt ata]
Starting 3/74  apport      [Running: anacron apt ata apport]
Starting 6/74  block       [Running: apt ata apport block]
Starting 7/74  boot        [Running: apt apport block boot]
```



Essential sos Command Examples

sos report

Purpose: Generates a full system diagnostic report (logs, configs, hardware info) as a .tar.xz archive.

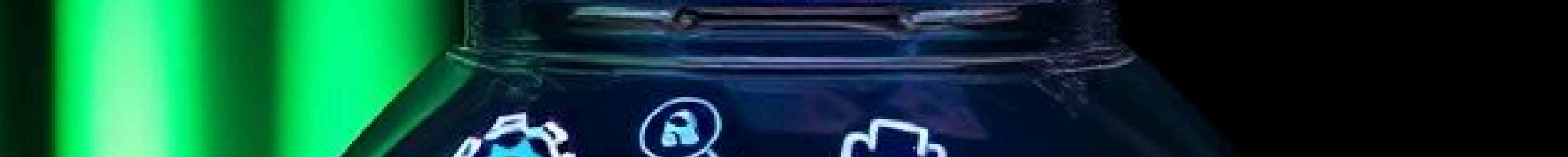
Use Case: Standard troubleshooting for support tickets.

sos report --list-plugins

Purpose: Lists available plugins to customize data collection.

Use Case: Identifying plugins (e.g., apache, kernel) for targeted reports.





Essential sos Command Examples

sos report --skip-plugins=plugin1,plugin2

Purpose: Excludes specific plugins to reduce report size or skip sensitive data.

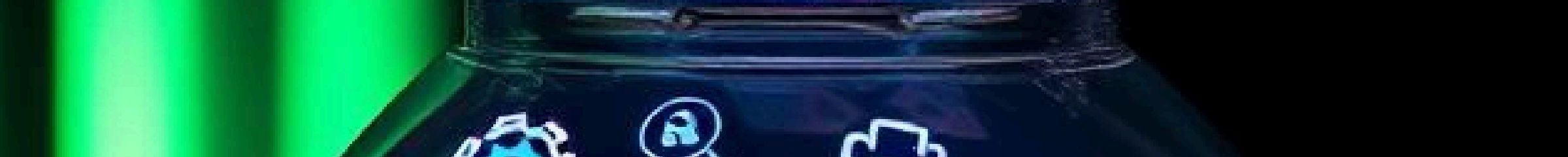
Use Case: Avoiding unnecessary or sensitive data collection.

sos report --enable-plugins=plugin1,plugin2

Purpose: Runs a report with specific plugins enabled for focused data collection.

Use Case: Collecting data only for specific services.





Why sos is Essential for SysAdmins

The sos command is a cornerstone for Linux troubleshooting, offering:

Comprehensive Diagnostics: Captures data from kernel, services, and hardware.

Time Efficiency: Automates collection of logs and configs, reducing manual effort.

Support Integration: Standardized format trusted by Red Hat, Canonical, and other vendors.

Use Case: When diagnosing a server crash, sos report provides logs (dmesg, syslog) and system state (e.g., ps, df) in one file for support analysis





```
🏠 | 📁 /tmp/sosreport-dell-laptop-2025-05-28-egndecb
ls -hl
Permissions Size User Date Modified Name
drwxr-xr-x - hosni 30 Apr 13:49 boot
lrwxrwxrwx - hosni 28 May 14:49 date -> sos_commands/systemd/timedatectl
lrwxrwxrwx - hosni 28 May 14:49 df -> sos_commands/filesys/df_al_x_autofs
lrwxrwxrwx - hosni 28 May 14:49 dmidecode -> sos_commands/hardware/dmidecode
-rw-r--r-- 232 hosni 28 May 14:40 environment
drwxr-xr-x - hosni 28 May 14:11 etc
lrwxrwxrwx - hosni 28 May 14:49 free -> sos_commands/memory/free
lrwxrwxrwx - hosni 28 May 14:49 hostname -> sos_commands/host/hostname
lrwxrwxrwx - hosni 28 May 14:49 installed-debs -> sos_commands/dpkg/dpkg_l
lrwxrwxrwx - hosni 28 May 14:49 ip_addr -> sos_commands/networking/ip_o_addr
lrwxrwxrwx - hosni 28 May 14:49 ip_route -> sos_commands/networking/ip_route_show_table_all
lrwxrwxrwx - hosni 28 May 14:49 last -> sos_commands/login/last_F
lrwxrwxrwx - hosni 28 May 14:49 lib -> usr/lib
lrwxrwxrwx - hosni 28 May 14:49 lsb-release -> sos_commands/release/lsb_release
lrwxrwxrwx - hosni 28 May 14:49 lsmod -> sos_commands/kernel/lsmod
lrwxrwxrwx - hosni 28 May 14:49 lsof -> sos_commands/process/lsof_M_n_l_c
lrwxrwxrwx - hosni 28 May 14:49 lspci -> sos_commands/pci/lspci_nnv
lrwxrwxrwx - hosni 28 May 14:49 mount -> sos_commands/filesys/mount_l
lrwxrwxrwx - hosni 28 May 14:49 netstat -> sos_commands/networking/netstat_W_neopa
drwxr-xr-x - hosni 24 May 16:38 opt
dr-xr-xr-x - hosni 28 May 13:12 proc
lrwxrwxrwx - hosni 28 May 14:49 ps -> sos_commands/process/ps_auxwwwm
lrwxrwxrwx - hosni 28 May 14:49 pstree -> sos_commands/process/pstree_lp
lrwxrwxrwx - hosni 28 May 14:49 root-symlinks -> sos_commands/host/find_-maxdepth_2_-type_l
drwxr-xr-x - hosni 28 May 14:39 run
drwxr-xr-x - hosni 28 May 14:40 sos_commands
drwxr-xr-x - hosni 28 May 14:40 sos_logs
drwxr-xr-x - hosni 28 May 14:40 sos_reports
dr-xr-xr-x - hosni 28 May 13:12 sys
lrwxrwxrwx - hosni 28 May 14:50 uname -> sos_commands/kernel/uname_a
lrwxrwxrwx - hosni 28 May 14:50 uptime -> sos_commands/host/uptime
```

